A High-Linearity Vernier Time-to-Digital Converter on FPGAs With Improved Resolution Using Bidirectional-Operating Vernier Delay Lines

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***Abstract*— Time-to-digital converters (TDCs) act as a core component in many timing-critical systems. Tapped delay line is the most widely used style for the field-programmable gate-array (FPGA)-based TDCs, but it suffers from several intractable problems such as poor nonlinearity and long delay-line occupation. Recently, a new ring-oscillator-based Vernier TDC circuit using carry chains was demonstrated to have much smaller nonlinearity with a much shorter delay line. Since the used delay line is not compensated, the timing jitter increases (or the precision degrades) remarkably when the oscillation number is large. This would incur undesirable lower resolution selection to maintain acceptable precision. In this article, a novel bidirectional-operating Vernier delay line structure is proposed to reduce the required oscillation number during the fine time measurement. Traditionally, for the Vernier delay line, the fast signal propagates along the slow delay line, while the slow signal propagates along the fast delay line, to which we call the normal Vernier delay line. The new structure proposes two parallelized Vernier delay line, consisting of one normal Vernier delay line and another abnormal Vernier delay line. The term "abnormal" refers to that the fast and slow signals are unusually fed to the fast and slow delay lines, respectively. A prototype TDC circuit based on this new method was implemented on a Stratix III FPGA. Test results demonstrate that by adopting the proposed novel circuit architecture, both the resolution and the root-mean-square error can be improved from the 30–40-ps level to the 20–30-ps level.**

***Index Terms*— Carry chain, field-programmable gate array (FPGA), time-to-digital converter (TDC), Vernier.**

I. Introduction

HIGH-RESOLUTION time-to-digital converters (TDCs) are needed in many high-energy nuclear experiments to help determine the particle type [1], [2]. TDCs can also be used in many other precise timing-related applications, such as medical time-of-flight positron emission tomography (PET) [3], [4], laser radar, automated test equipment (ATE), and all-digital phase locked loop (PLL) circuit [5].

Field-programmable gate-array (FPGA)-based TDCs own the unique benefits of low cost, reconfigurability, and flexibility, and fast developing period, when compared with their application-specific integrated-circuit (ASIC) counterparts. The TDCs usually contain two time measurement steps: one coarse time module to enlarge the entire measurement range and one fine time module to improve the resolution. The fine time module should measure the time range covering the whole period of the coarse time module. The coarse time counter (given by the coarse time module) and fine time counter (given by the fine time module) combined together generates a timestamp, representing the moment when an event occurs. Two such TDC channels can be built to measure a fine time interval.

For the FPGA platforms, the dominant technique used to build the fine time module is the tapped delay line (TDL) [6]–[21]. Numerous works have demonstrated that carry chains, which are fabricated by FPGA manufacturers to support fast arithmetic calculations, are very powerful tools to construct TDLs. One benefit is that the time delay between the successive carry chain units is very small, with modern ones being smaller than 10 ps. Another benefit is that the carry chains are always accompanied by their following register chains in a modern FPGA fabric, which is convenient for the software compiler to generate the TDL automatically. However, there are some intractable problems for the TDL-based TDCs. The ultrawide bin (which usually occurs when the carry chain spans across the adjacent logic blocks) can significantly deteriorate the resolution, differential nonlinearity (DNL) error, and integral nonlinearity (INL) error. To construct a high-resolution TDL, the length of the carry chain has to be rather long, being at least longer than 100 delay units in modern designs, which would incur large resource cost. To overcome the above problems, Wu and Shi [22] proposed the wave union method, whose core idea is to measure a time event several times and average the results to mitigate the influence of the ultrawide bins. The multichannel averaging method by constructing several parallel TDLs in space has similar performance improvement, but requires more resource

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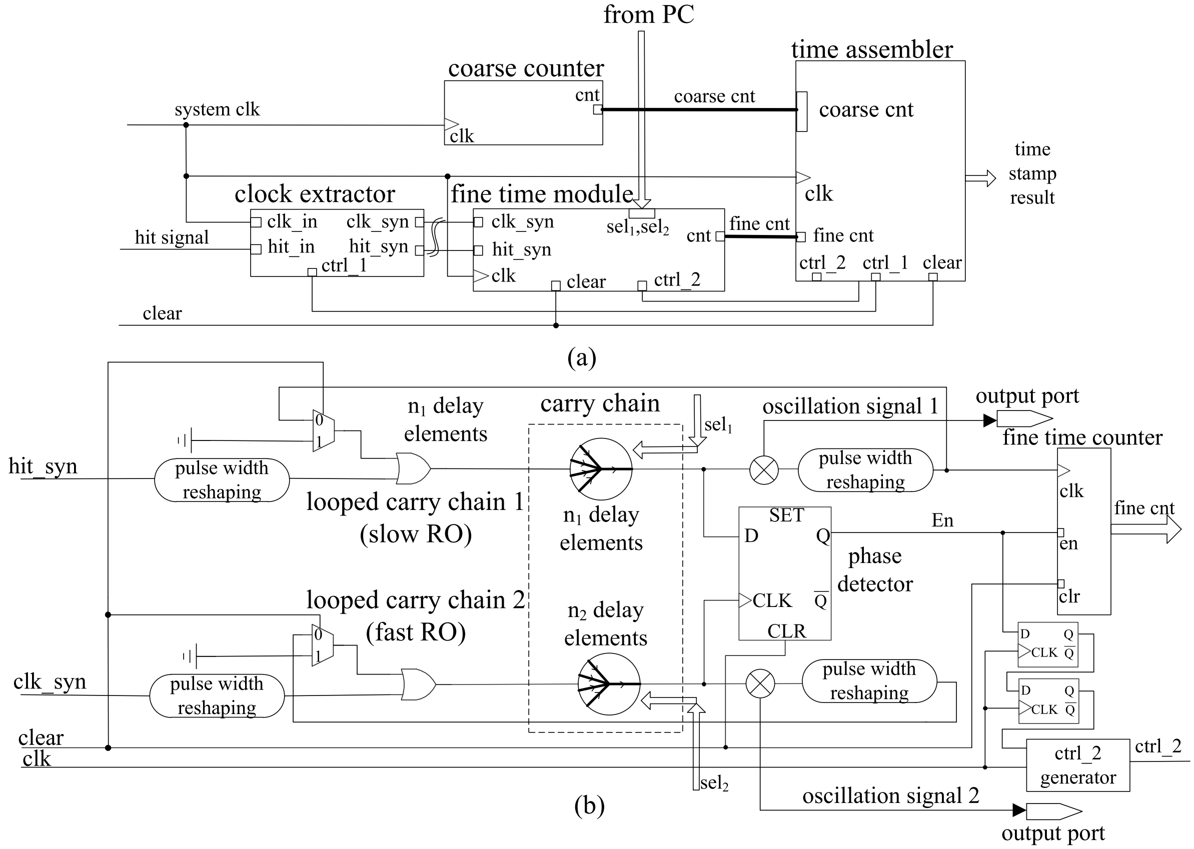

Fig. 1. RO-based Vernier TDC circuit architecture. (a) Overall structure. (b) Specific structure of the fine time module.

cost [17]. Other methods to improve resolution and reduce DNL include bin-by-bin calibration [8], bin decimation [18], and TDL combined with phased clocks [19], [20]. Using these optimization methods, TDCs can already obtain high resolution below 10 ps. However, the obtained DNL still lies in the range of 2–4 least significant bits (LSBs).

Recently, our group proposed to organize carry chains in looped style to form ring oscillators (ROs) [23], [24]. Two ROs are built to work in the Vernier mode, and the resolution is determined by their oscillation period difference. This kind of TDC owns much smaller DNL and INL (mostly smaller than 1 LSB), since they are not dependent on the ultrawide bin any more. It requires shorter delay line, so it costs much less resource. Although it also has the potential to obtain sub-10-ps resolution by adjusting the oscillation periods, the accumulated oscillation jitter ultimately limits the achievable resolution. Since the delay of the ROs is not compensated, the root-mean-square (rms) error increases proportionally with the square root of the oscillation number [25]. Therefore, the resolution cannot be chosen too small to avoid a large rms error. To solve this problem, a 2-D Vernier delay line may be a useful technique. It reduces the total oscillation number by assigning different delay values for each delay elements along the fast and slow delay lines [26]–[28]. However, this technique is not friendly to the FPGA-based designs, since the delay of the basic delay elements is not easy to be controlled as those in the ASIC-based designs.

In this article, a novel bidirectional-operating Vernier delay line structure is proposed to reduce the required oscillation number in the fine time measurement. It consists of two parallelized delay lines with one being the normal Vernier delay line and the other one being the abnormal Vernier delay line. One can imagine that when an event triggers the two different types of Vernier delay line simultaneously, it generates two numerically opposite fine time values. When one value is big, the other is small. After synthesizing them by choosing the smaller value, the oscillation number of the TDC can be reduced by a fact of 2, which is beneficial to improve the precision. Or equivalently, for the same precision target, the resolution can be selected much higher to tolerate two times larger oscillation number of the RO. Note that the period difference is originally modified by the manual intervention and recompilation processes [23], but here, we use a newly proposed dynamically adjustable delay RO structure, which uses a multiplexer to select the actual RO length in run time to control the resolution [29]. This can greatly save the TDC design time.

II. Design

A. Circuit Architecture

One channel of the TDC architecture is shown in Fig. 1. It contains two steps: one coarse time module and one fine time module. The coarse time module uses a counter to measure the elapsed time at a coarser level, which is determined by the adopted clock rate. The fine time module uses the RO-based

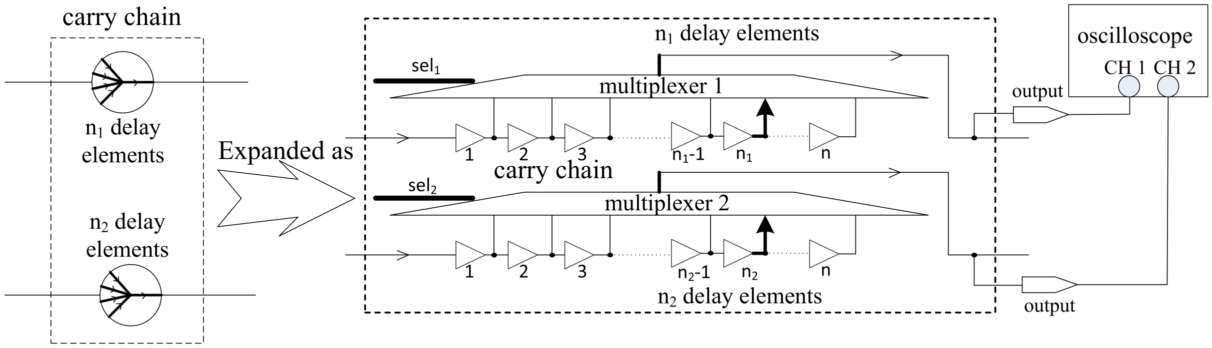

Fig. 2. Dynamically delay changeable RO structure.

Vernier delay line to interpolate finely the clock period of the coarse counter to improve the overall measurement accuracy. A complete Vernier delay line contains two ROs with different oscillation periods. The resolution of the TDC is determined by the oscillation-period difference in the two ROs. The clock extractor module in Fig. 1(a) extracts the nearest clock signal after the hit signal (event signal) to the fine time module. The fine time module measures the time interval between the hit signal and the extracted clock signal. The time assembler module combines the coarse counter and fine counter values to generate the timestamp. The delay configuration parameters are sent from PC to control the actual length of the two ROs.

The implementation details of the fine time module are depicted in Fig. 1(b). The ROs are formed by connecting the last delay unit to the first delay unit. By lending the advantage of the fast propagation attribute of the carry chain and assigning different delay unit numbers to each RO, their oscillation period difference can be controlled very small. The upper slow RO receives the leading hit signal, and the bottom fast RO receives the lagging clock signal. This is the normal working mode for the Vernier delay line. The leading hit_syn signal can also be unusually fed to the fast RO, and the lagging clk_syn signal to the slow RO. This is the abnormal working mode for the Vernier delay line. We will show in Section II-B that by synthesizing the fine time results given by the normal and abnormal Vernier delay lines, the maximal oscillation number can be reduced by a factor of 2, thus reducing the rms error and improving the resolution of the TDC.

The specific structure of the dynamically delay changeable RO is depicted in Fig. 2. An n-input multiplexer is placed alongside the looped carry chain, and each of the delay units is connected to one input of the multiplexer. The RO can contain any delay unit number according to the specific channel switch parameter. If the looped carry chain contains more delay units, it will have a larger oscillation period. The actual length of each RO is dynamically configured by the PC from the USB communication module, so the resolution of the TDC can be optimized by trying different length combinations. To observe the actual oscillation periods, the oscillation signals are brought out using an external oscilloscope.

The structure of another key module, the clock extractor, is depicted in Fig. 3. It retrieves the fine time between the clock and hit signals for the following fine time module. The working principle of the clock extractor belongs to some kind

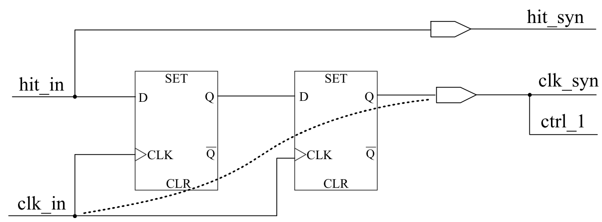

Fig. 3. Clock extractor structure.

of phase detector, so it uses the D flip-flop to fulfill the task. Two cascaded D flip-flops are adopted to avoid metastability problem when the hit and clock signals are too close to each other.

B. Bidirectional-Operating Vernier Delay Line Method

The rms error σ increases proportionally to the square root of the total oscillation number n for an uncompensated RO. The relation is expressed by the following equation [23]:

$$\sigma = k \cdot \sqrt{n \cdot T_{cyc}} = k \cdot \sqrt{\frac{\Delta T}{r} \cdot T_{cyc}} \tag{1}$$

where k is a circuit-dependent constant factor that is mainly influenced by the overall circuit noise, T_{cyc} represents the oscillation period of the ROs, n represents the oscillation number, r represents the resolution of the fine time module, and ΔT represents the fine time interval that needs to be measured. It should be noted that the worst rms error occurs when the fine time module measures the longest time interval $\Delta T = T_{clk}$, where T_{clk} represents the clock period of the coarse time module. Equation (1) denotes that the resolution cannot be set too small to avoid too a large rms error. For a typical 5–8-ns RO on the FPGA, a good-resolution choice turns out to be about 30–40 ps [23], with which the rms error can be constrained not larger than the resolution to guarantee reasonable accuracy.

Equation (1) gives the clue that if the maximal oscillation number can be reduced, the obtained rms error and resolution can be improved accordingly. This article proposes a novel circuit implementation architecture, as depicted in Fig. 4(a), to realize this goal. Two parallelized Vernier delay lines with opposite working modes are constructed. The time synthesizer module after the two TDC channels is designed to accept the two timestamp results and sort out the one with the smaller

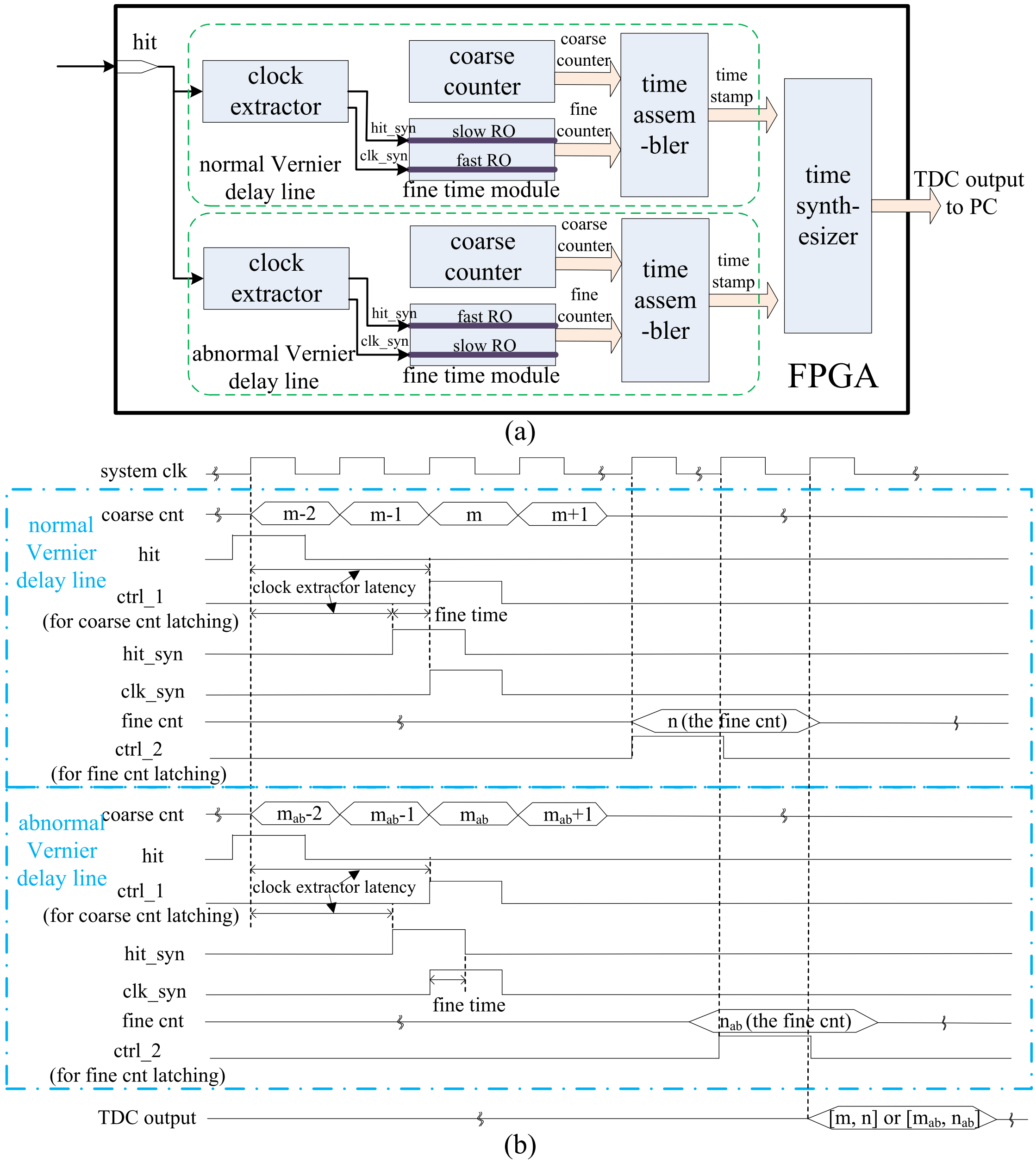

Fig. 4. Bidirectional-operating Vernier delay line structure. (a) Circuit architecture. (b) Timing diagram.

value. Fig. 4(b) shows the timing diagram of the proposed TDC architecture. The ctrl_1 signal denotes the correct timing to latch the coarse counter, and the ctrl_2 signal denotes the correct timing to latch the fine counter. The two Vernier delay lines work in parallel and give two opposite results. The time synthesizer module gives the final result by analyzing the values of the fine counter. For the normal Vernier TDC channel, the TDC output is the same as that of the TDC channel output. However, for the abnormal Vernier TDC channel, the fine time counter value of the TDC output is recalculated as $n = \left\lceil \frac{T_{clk}}{r_{ab}} \right\rceil - n_{ab}$, where the sublabel "$ab$" denotes that the parameter is for the abnormal Vernier TDC channel.

After the two signals are fed to their corresponding delay lines, the clk_syn signal uses a D flip-flop [see Fig. 1(b)] as the phase detector to sample the state of the hit_syn signal. As shown in the left part of Fig. 5 for the normal Vernier mode, the sampling result is high in the initial state, and it will turn low after the clk_syn signal surpasses the hit_syn signal. The clk_syn signal propagates toward the front end of the hit_syn signal in this mode. While in the abnormal mode, as shown in the right part of Fig. 5, the clk_syn signal propagates toward the rear end of the hit_syn signal. The sampling signal in this case changes from the initial high level to the final low level when the positive duration of the hit_syn signal completely surpasses the clk_syn signal. When the clk_syn signal falls into the first-half positive duration of the hit_syn signal, the normal Vernier delay line gives the smaller oscillation number. Otherwise, when the clk_syn signal falls into the second-half positive duration of the hit_syn signal, the abnormal Vernier delay line gives the smaller oscillation number. Especially, when the clk_syn signal falls exactly to the half-positive position of the hit signal, the two Vernier delay

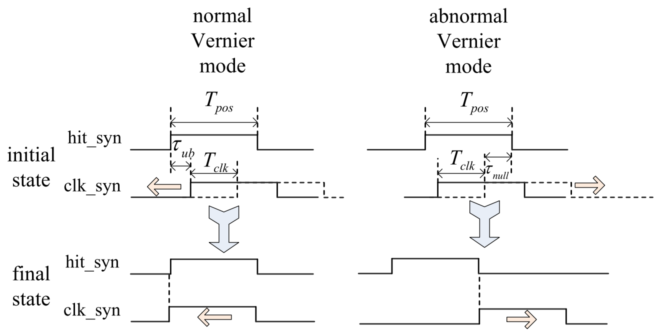

Fig. 5. Working principle of the bidirectional-operating Vernier delay lines.

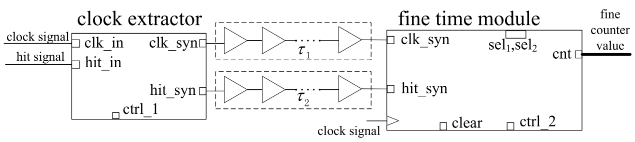

Fig. 6. Inserted delay chains of cascaded inverters.

lines give the same oscillation number, which is the maximal value under this circuit architecture. The maximal oscillation can be reduced by a factor of 2 compared with the original TDC design, so the resolution can be made higher according to (1).

C. Implementation Process

It should be noted that the above analysis is obtained by requiring that $T_{pos} \geq T_{clk}$ and the timing parameters τ_{ub} and τ_{null} in Fig. 5 to be 0, where T_{pos} represents the positive duration of the hit_syn signal, τ_{ub} represents the additional delay between the front ends of the hit_syn and clk_syn signals when $\varDelta T = 0$, and τ_{null} represents the additional delay between the rear ends of the hit_syn and clk_syn signals when $\varDelta T = T_{clk}$. As shown in Fig. 5, the clk_syn signal waggles for the T_{clk} duration relative to the hit_syn signal. The condition $T_{pos} \geq T_{clk}$ guarantees that, for each $\varDelta T$, the positive duration of the hit_syn signal can enclose the arrival time of the clk_syn signal, so that the D flip-flop phase detector can work correctly in the initially high-level status. The τ_{ub} originates from the unbalanced delay path between the hit_syn and clk_syn signals from the clock extractor module to the fine time module. Ideally, τ_{ub} should be 0, but practically, it is not easy to be rigidly satisfied. The positive nonzero τ_{ub} will increase the oscillation number for the normal Vernier delay line and deteriorate the rms error performance. Note that τ_{ub} should not be negative, since in that case, for the small fine time interval $\varDelta T < |\tau_{ub}|$, the normal Vernier delay line cannot function correctly.

The τ_{null} is always positive when $T_{pos} > T_{clk}$. If the condition $T_{pos} = T_{clk}$ can be ideally satisfied, the τ_{null} becomes 0. However, in the actual circuit implementation, the condition is also not easy to be rigidly satisfied. The nonzero τ_{null} results in the oscillation number increase for the abnormal Vernier delay line, and correspondingly the rms error performance deterioration.

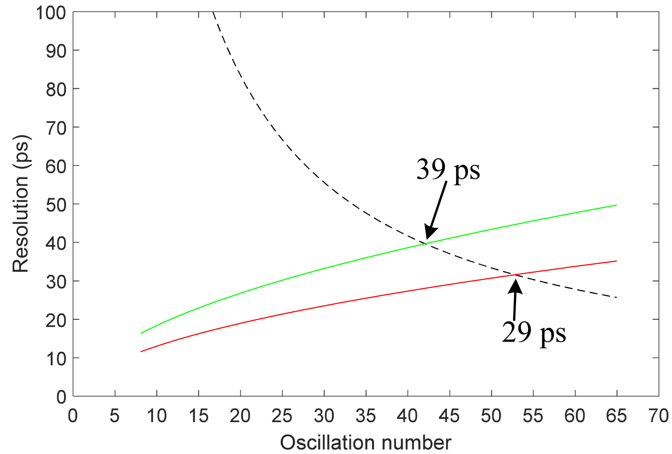

Fig. 7. Empirical rms error curve and the resolution curve.

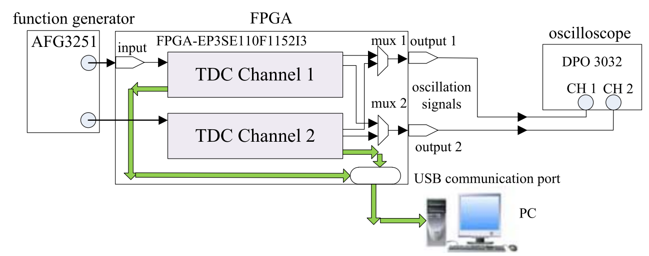

Fig. 8. Experimental setup.

In order to overcome the above problems, two chains of the cascaded lookup table (LUT)-implemented inverters are inserted between the clock extractor module and the fine time module for each of the ROs to adjust τ_{null} and τ_{ub} in the TDC design (see Fig. 6). The inverter amount is totally 32 but is actually modified to let both τ_{null} and τ_{ub} positive and as small as possible (see the following step 2). The τ_{null} and τ_{ub} are not easy to be measured directly, but their values can be inferred from the distribution of the fine time counter value (or oscillation number; see Fig. 9). The smallest counter number should be adjusted larger than 0 and as small as possible by modifying the delay chain length.

Based on the above analysis, the design process is summarized as follows.

1) TDC circuit architecture depicted in Figs. 1–3 is written using the Verilog language and automatically complied by software. To guarantee that all ROs have close oscillation periods in order to facilitate the resolution-determination process, the two-step design method for the fine time module is adopted [23]. A prototype RO is first designed in an independent project, and its fitting result is exported out as the seed RO. Then, in the TDC circuit design, all the ROs are imported from the same seed RO fitting result, so they have identical structure after software compilation. This makes the following resolution adjustment process much easier.
2) Modifying the inverter number for each of the delay chains lying between the clock extractor module and the fine time module to let both τ_{ub} (for the normal Vernier mode delay line) and τ_{null} (for the abnormal

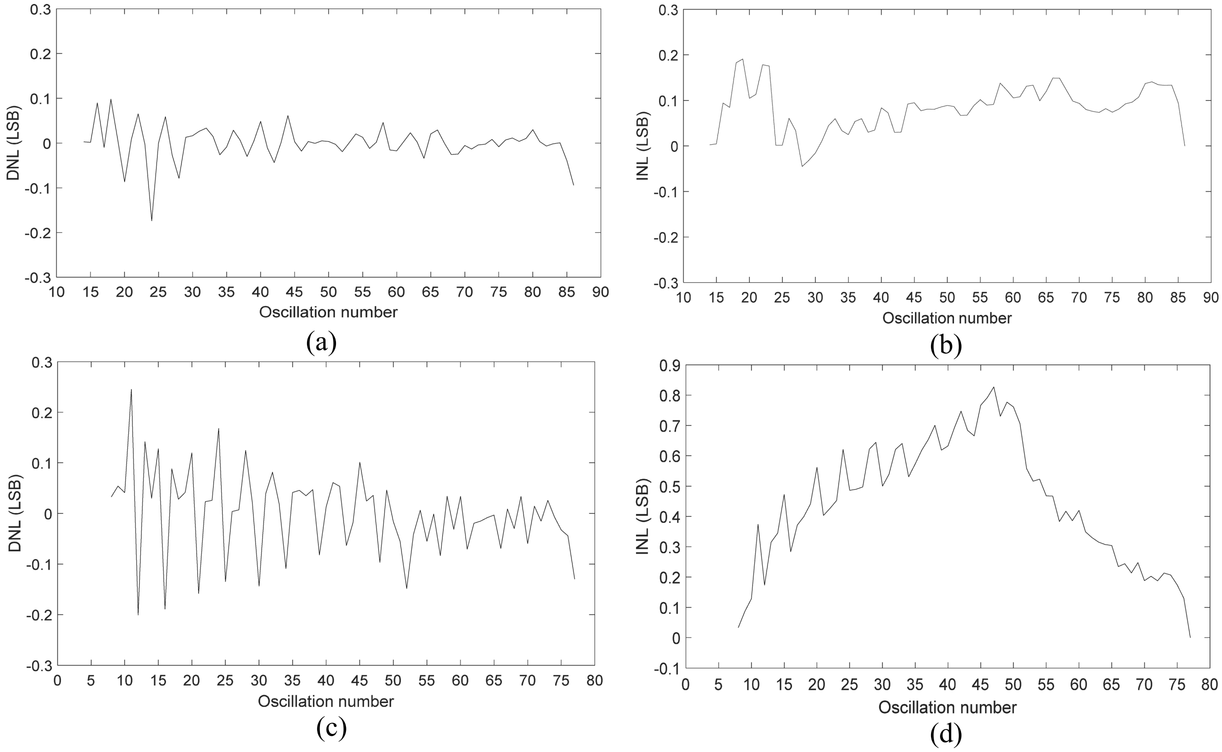

Fig. 9. DNL and INL curves of the TDC channel No. 1. (a) DNL for the normal Vernier delay line. (b) INL for the normal Vernier delay line. (c) DNL for the abnormal Vernier delay line. (d) INL for the abnormal Vernier delay line.

Vernier mode delay line) positive and as small as possible. This step needs manual intervention to the TDC circuit fitting result by utilizing some resource-editing tool (the engineering change order (ECO) tool for the Altera software and the FPGA editor tool for the Xilinx software). For example, in the Altera FPGA design, by using the chip planner tool, the final inverter of the delay chain can be physically located. Then, by using the ECO tool, the input port of the final inverter can be disconnected from its previous one and reconnected to any other former inverter with a smaller index, by doing which the actual length of the delay chain is shortened. Although recompilation is needed in the process, this step is very easy to be fulfilled.

3) Dynamically configure different channel selection parameter combinations for all the Vernier delay lines. For any parameter combination, their oscillation period can be observed from an external oscilloscope, so the resolution can be evaluated in real time. The parameter combination satisfying the design target is sorted out as the configuration parameter.

The above TDC circuit contains one normal Vernier delay line and one abnormal Vernier delay line. Since each Vernier delay line contains two ROs, one such TDC channel contains four ROs.

Finally, the resolution design target is discussed. The experimental rms error versus oscillation number curve for the RO having about 7-ns oscillation period (corresponding to the implementation case in this article) is depicted in Fig. 7 (the green solid line) [23]. The resolution of the TDC versus the maximal oscillation number in the mathematical form of $r = \frac{T_{clk}}{n}$ is depicted in the black dashed line. The T_{clk} is 1667 ps corresponding to the used 600-MHz clock rate in our design. The two curve intersects at the point P_1, which denotes the oscillation number that makes the rms error equaling the resolution. A larger oscillation number makes higher resolution but poorer rms error. The resolution corresponding to P_1 point was chosen as the design target in the previous work, which was 39 ps. In the new bidirectional operation Vernier delay line circuit implementation, the effective rms error is reduced by a factor of $\sqrt{2}$, whose curve is depicted in red solid line in Fig. 7. The resolution curve intersects with the red line at point P_2, which gives the improved target value of about 29 ps. Based on the above analysis, our resolution design target is set to 20–30 ps in this article.

III. Experimental Results

The proposed TDC architecture was implemented on an Intel Stratix III FPGA chip. Two of such TDC channels containing eight ROs are constructed to be able to measure the time interval. In our design, the coarse time counter has a 9-bit width and runs under the clock rate of 600 MHz (corresponding to $T_{clk} = 1667$ ps). The fine counter has a 7-bit width to support high enough resolution (supporting up to $\frac{1667}{2^7} = 13$ ps). The hit signals are generated from an arbitrary function generator (AFG3251 by Tektronix Co.) with a repetition rate of 200.1 KHz. The internal oscillation signals are connected to an external oscilloscope with the series number of DPO 3032 from Tektronix Co. to assist the period adjustment. The adopted experimental setup is depicted

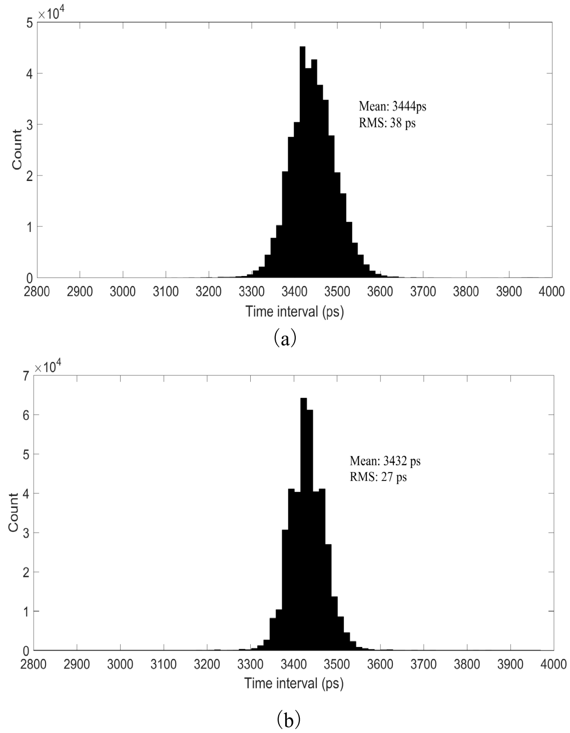

Fig. 10. Histogram comparison of a small time interval of about 3 ns between (a) original result and (b) revised result.

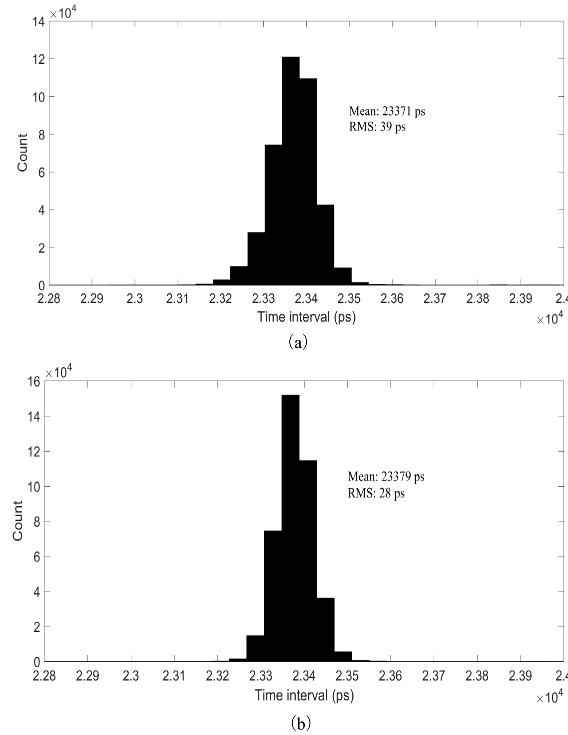

Fig. 11. Histogram comparison of a large time interval of about 23 ns between (a) original result and (b) revised result.

in Fig. 8. Since the frequency of the hit signal is asynchronous to the frequency of the clock of the constructed TDCs, code density test can be applied to determine the resolution, and the DNL and INL performance. All the following test results are obtained under the nominal voltage supply of 1100 mV and the ambient temperature of 20 °C.

First, to evaluate the resolution, and the DNL and INL performance, all fine time counter results should be uploaded to PC. The time synthesizer module was forbidden in the test mode to upload directly all the needed data. Fig. 9 depicts the obtained DNL and INL curves of the TDC channel No. 1. It can be observed that for the normal Vernier delay line, the DNL lies in the range of −0.17–0.10 LSB, and the INL lies in the range of −0.04–0.19 LSB. For the abnormal Vernier delay line, the DNL lies in the range of −0.20–0.25 LSB, and the INL lies in the range of 0.03–0.82 LSB. All results are relatively small compared with the conventional TDL-based design. Since the bin number range for the normal (abnormal) Vernier delay line is 14–86 (8–77), the corresponding resolution is $\frac{1667}{68} = 24.5$ ps ($\frac{1667}{69} = 24.2$ ps). Similar performance is obtained for the TDC channel No. 2, whose DNL lies in the range of −0.16–0.22 LSB (−0.09–0.31 LSB), INL lies in the range of −0.21–0.17 LSB (−0.11–0.45 LSB) for the normal (abnormal) Vernier delay line, and resolution is 23.2 ps (22.8 ps) for the normal (abnormal) Vernier delay line.

Fine time-interval test was then applied to evaluate the rms error performance. A shorter fine time of 5 ns and a longer fine time of 25 ns were generated by the function generator to carry on the tests. The time information of the TDC channel No. 2 is subtracted from that of the TDC Channel No. 1 to calculate the measured fine time interval. To reduce the statistical error influence, more than 500 000 data are accumulated for each measurement case. To demonstrate that the proposed bidirectional operation Vernier delay line is beneficial to decreasing the rms error, the result calculated by using only the normal Vernier delay lines (called the original result) is compared with that calculated by choosing the Vernier delay line that holds smaller fine time counter (called the revised result). Fig. 10 depicts the histograms of the time distribution with a small interval of about 3 ns. The original rms error result is 38 ps, while the revised result is 27 ps, which shows a remarkable improvement. Fig. 11 depicts the histograms of the time distribution with a large interval of about 23 ns. Similarly, the calculated rms error result is improved from the original value of 39 ps to the revised value of 28 ps. It can be seen that due to the adoption of the proposed novel circuit architecture, both resolution and rms error of the RO-based TDC can be improved from the 30–40 ps-level [23] to the 20–30-ps level. No calibration circuit is involved in this design, since the adopted Vernier structure makes the drifting coefficients versus temperature and (or) voltage change rather small (about 0.1 ps/°C and −0.2 ps/mV) [29]. A time deviation of about −1.6 ns (3.4–5 and 23.4–25 ns) from the configured value of the function generator is observed in both tests. This is mainly caused by the unequal delays of the input pins of the two TDC channels.

Compilation reports give the following resource cost: LUTs being 986 and registers being 172, which shows small resource requirement. Power simulation gives the estimated power consumption of 534 mW. The main drawback of such TDC is its longer dead time. The dead time is calculated as $T_{cyc} \cdot n_{max} = 7 \cdot 86 = 602$ ns, where n_{max} represents

TABLE I
PERFORMANCE COMPARISONS BETWEEN THIS ARTICLE AND SOME OTHER RECENT FPGA-BASED WORKS

ref.	chip	method	resolution (ps)	precision RMS (ps)	DNL (LSB)	INL (LSB)	dead time (ns)	registers cost	LUTs cost
[10]	Virtex-6	TDL+phased clock	10	12.8	-1 ~ 1.91	-2.2 ~ 3.93	N/S	N/S	N/S
[16]	Virtex-5	TDL	30	15	-1 ~ 3	-4 ~ 4	30	571	1064
[30]	UltraScale	TDL+dual sampling	4.5	3.9	N/S	N/S	4	N/S	N/S
[31]	Cyclone II	TDL	21.8	28.8	N/S	N/S	N/S	23494	28085
[22]	Cyclone II	TDL+wave union	2.44	10	N/S	N/S	45	N/S	6851
[32]	Spartan-3	pulse shrinking	42	56	-0.98 ~ 0.6	-4.17 ~ 3.6	710	N/S	N/S
[19]	Spartan-6	TDL+ multi-edge coding	0.9	<6	<2.91	<15.70	3	N/S	144 SLICEs
[20]	Kintex-7	TDL+phased clock +time coding line	2	4.5	<3.84	<20.9	11	N/S	N/S
[23]	Stratix III	RO-based Vernier	31	36.1	-0.080 ~ 0.073	-0.087 ~ 0.091	256	319	104
[24]	Stratix III	RO-based Vernier	37	39	-0.4 ~ 0.4	-0.7 ~ 0.7	400	319	104
this work	Stratix III	RO-based Vernier+ Bidirectional-Operating	24.5	28	-0.20 ~ 0.25	0.03 ~ 0.82	602	986	172

N/S=not specified

the maximal fine time counter. Performance comparisons between this article and some other recently state-of-the-art FPGA-based TDC designs are summarized in Table I.

IV. CONCLUSION

This article presents a novel technique called the bidirectional operation Vernier delay line to decrease the required oscillation number during the fine time measurement process. This can remarkably decrease the rms error and improve the resolution. Our prototype circuit implementation on the Stratix III FPGA demonstrates that the obtained resolution can be improved from the 30–40-ps level to the 20–30-ps level. Combining other unique advantages such as low nonlinearity error and low resource cost, we believe that the RO-based Vernier TDCs can play an important role in any practical fine time measurement tasks.

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